

Statistics 651

Multivariate Models

Winter 2026

Objectives

1. Work with common multivariate distributions
2. Build (conditionally) conjugate models for multinomial and Multivariate Normal data

Resources

- BDA3:
 - Ch 3.4 (Multinomial, Dirichlet)
 - Ch 3.5-3.6 (Multivariate Normal)
- Hoff: Ch 7 (Multivariate Normal)
- [Rencher and Schaalje](#), Ch 4 (Multivariate Normal theory)

Multivariate Normal

Multivariate Normal

$$\underline{a}^T \Sigma \underline{a} > 0$$

i Definition

Let $\boldsymbol{\mu} \in \mathbb{R}^D$ and $\boldsymbol{\Sigma}$ be a $D \times D$ positive-definite matrix.

The **Multivariate Normal** or (Gaussian or MVN) density for $\boldsymbol{x} = (x_1, \dots, x_D) \in \mathbb{R}^D$ is

$$f(\boldsymbol{x}) = (2\pi)^{-D/2} \det(\boldsymbol{\Sigma})^{-1/2} \exp \left[-\frac{1}{2} (\boldsymbol{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\boldsymbol{x} - \boldsymbol{\mu}) \right].$$

We write $\boldsymbol{x} \sim \text{Normal}_D(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

- $\mathbb{E}(\boldsymbol{x}) = \boldsymbol{\mu}$
- $\text{Cov}(x_i, x_j) = (\boldsymbol{\Sigma})_{i,j}$
- $\boldsymbol{\Sigma}^{-1}$ is also positive definite and is called the **precision matrix**.
- Contours of the density are elliptical.



Multivariate Normal

Let $\mathbf{x} \in \mathbb{R}^{D_x}$, $\mathbf{y} \in \mathbb{R}^{D_y}$, and

$$\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} \sim \text{Normal}_{D_x+D_y}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \text{ with } \boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_x \\ \boldsymbol{\mu}_y \end{pmatrix} \text{ and } \boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{xx} & \boldsymbol{\Sigma}_{xy} \\ \boldsymbol{\Sigma}_{yx} & \boldsymbol{\Sigma}_{yy} \end{pmatrix}.$$

Marginal distributions

The marginal distributions of subvectors are also normal:

$$\mathbf{x} \sim \text{Normal}_{D_x}(\boldsymbol{\mu}_x, \boldsymbol{\Sigma}_{xx}),$$

$$\mathbf{y} \sim \text{Normal}_{D_y}(\boldsymbol{\mu}_y, \boldsymbol{\Sigma}_{yy}).$$

Note that two marginally normal random vectors are NOT necessarily jointly normally distributed.

Multivariate Normal

Let $\mathbf{x} \in \mathbb{R}^{D_x}$, $\mathbf{y} \in \mathbb{R}^{D_y}$, and

$$\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} \sim \text{Normal}_{D_x+D_y}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \text{ with } \boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_x \\ \boldsymbol{\mu}_y \end{pmatrix} \text{ and } \boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{xx} & \boldsymbol{\Sigma}_{xy} \\ \boldsymbol{\Sigma}_{yx} & \boldsymbol{\Sigma}_{yy} \end{pmatrix}.$$

Conditional distributions

The conditional distributions of subvectors are also normal:

$$\mathbf{x} \mid \mathbf{y} \sim \text{Normal}_{D_x}(\boldsymbol{\mu}_{x|y}(\mathbf{y}), \boldsymbol{\Sigma}_{x|y}),$$

$$\mathbf{y} \mid \mathbf{x} \sim \text{Normal}_{D_y}(\boldsymbol{\mu}_{y|x}(\mathbf{x}), \boldsymbol{\Sigma}_{y|x}),$$

where

$$\boldsymbol{\mu}_{y|x}(\mathbf{x}) = \boldsymbol{\mu}_y + \boldsymbol{\Sigma}_{yx} \boldsymbol{\Sigma}_{xx}^{-1}(\mathbf{x} - \boldsymbol{\mu}_x)$$

and

$$\boldsymbol{\Sigma}_{y|x} = \boldsymbol{\Sigma}_{yy} - \boldsymbol{\Sigma}_{yx} \boldsymbol{\Sigma}_{xx}^{-1} \boldsymbol{\Sigma}_{xy}.$$

Note that $\mathbf{x}$ and $\mathbf{y}$ are independent if and only if $\boldsymbol{\Sigma}_{xy} = \boldsymbol{\Sigma}_{yx}^\top$ contains all 0s.

Multivariate Normal

Linear transformations of normal vectors

Let $\mathbf{x} \sim \text{Normal}_D(\boldsymbol{\mu}, \boldsymbol{\Sigma})$,

$\mathbf{c} \in \mathbb{R}^N$, and $\mathbf{A} \in \mathbb{R}^{N \times D}$.

Then

$$\mathbf{c} + \mathbf{A}\mathbf{x} \sim \text{Normal}_N(\mathbf{c} + \mathbf{A}\boldsymbol{\mu}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top).$$

Multivariate Normal

The linear properties of the normal distribution lead to *easy* marginalization (without calculus).

Proposition

If $\mathbf{x} \sim \text{Normal}_{D_x}(\boldsymbol{\mu}_x, \boldsymbol{\Sigma}_x)$ independent of $\mathbf{y} \sim \text{Normal}_{D_y}(\boldsymbol{\mu}_y, \boldsymbol{\Sigma}_y)$, then for $\mathbf{c} \in \mathbb{R}^N$ $\mathbf{A} \in \mathbb{R}^{N \times D_x}$ and $\mathbf{B} \in \mathbb{R}^{N \times D_y}$,

$$\mathbf{c} + \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{y} \sim \text{Normal}_N(\tilde{\boldsymbol{\mu}}, \tilde{\boldsymbol{\Sigma}})$$

where $\tilde{\mu} = E[C + Ax + By] = C + AE(x) + BE(y)$
and $\tilde{\Sigma} =$

and $\tilde{\Sigma} =$

$$A \Sigma_x A^T + B \Sigma_y B^T$$

Proof: MGF, or $\begin{pmatrix} x \\ y \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix}, \begin{pmatrix} \Sigma_x & 0 \\ 0 & \Sigma_y \end{pmatrix}\right)$

Multivariate Normal

💡 Example (marginalization trick)

Let

$$\begin{aligned}\boldsymbol{\mu} &\sim \text{Normal}_D(\mathbf{m}, \mathbf{V}), \\ \mathbf{y} \mid \boldsymbol{\mu} &\sim \text{Normal}_D(\boldsymbol{\mu}, \boldsymbol{\Sigma}).\end{aligned}$$

Find the marginal distribution of $\mathbf{y}$.

$$P(\mathbf{y}) = \int P(\mathbf{y} \mid \boldsymbol{\mu}) d\boldsymbol{\mu} = \int p(\mathbf{y} \mid \boldsymbol{\mu}) p(\boldsymbol{\mu}) d\boldsymbol{\mu}$$

$$\boldsymbol{\mu} = \mathbf{m} + \boldsymbol{\varepsilon}_0 \quad \text{where } \boldsymbol{\varepsilon}_0 \sim N(\mathbf{0}, \mathbf{V})$$

$$\begin{aligned}\mathbf{y} &= \boldsymbol{\mu} + \boldsymbol{\varepsilon} \quad \text{where } \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \boldsymbol{\Sigma}) \\ &\quad \text{and } \text{Cov}(\boldsymbol{\varepsilon}_0, \boldsymbol{\varepsilon}) = \mathbf{0}\end{aligned}$$

$$= \mathbf{m} + \boldsymbol{\varepsilon}_0 + \boldsymbol{\varepsilon}$$

$$E(\mathbf{y}) = \mathbf{m} + \mathbf{0} + \mathbf{0} \quad \text{Cov}(\mathbf{y}) = \text{Cov}(\mathbf{m} + \boldsymbol{\varepsilon}_0 + \boldsymbol{\varepsilon}, \mathbf{m} + \boldsymbol{\varepsilon}_0 + \boldsymbol{\varepsilon})$$

$$\begin{aligned}
&= \cancel{\text{Cov}(m, m)} + \cancel{\text{Cov}(m, \varepsilon_0)} + \\
&\quad \cancel{\text{Cov}(m, \varepsilon)} + \cancel{\text{Cov}(\varepsilon_0, m)} + \\
&\quad \text{Cov}(\varepsilon_0, \varepsilon_0) + \cancel{\text{Cov}(\varepsilon_0, \varepsilon)} + \\
&\quad 0 + 0 + \text{Cov}(\varepsilon, \varepsilon) \\
&= V + \Sigma
\end{aligned}$$

So marginally, $y \sim \mathcal{N}_o(\underline{m}, V + \Sigma)$

Square root matrix

If Σ is positive definite, how might we find a “square root” matrix?

Two paths:

1. Diagonalization
2. Cholesky factorization

Square root matrix

Diagonalization

If Σ is positive definite, we can perform a **spectral decomposition**

$$\Sigma = QDQ^\top$$

where D is diagonal with positive elements and $Q^\top Q = I$.

Then

$$\Sigma^{1/2} = QD^{1/2}Q^\top.$$

Proof:

$$\begin{aligned}\Sigma^{1/2} \Sigma^{1/2} &= QD^{1/2}Q^\top QD^{1/2}Q^\top \\ &= QD^{1/2} \underbrace{Q^\top Q}_I D^{1/2} Q^\top \\ &= QDQ^\top\end{aligned}$$

Square root matrix

Definition

A **Cholesky factor** of positive definite matrix Σ is a square, lower-triangular matrix $\mathbf{L}$ such that

$$\mathbf{L}\mathbf{L}^\top = \Sigma.$$

$$\mathbf{L} = \begin{pmatrix} L_{11} & 0 & 0 & 0 \\ L_{21} & L_{22} & 0 & 0 \\ L_{31} & & \ddots & 0 \\ L_{D1} & & & L_{DD} \end{pmatrix}$$

See [Cholesky factorization Wikipedia page](#).

Example

Let $\mathbf{z} \sim \text{Normal}_D(\mathbf{0}, \mathbf{I})$.

What is the distribution of $\boldsymbol{\mu} + \mathbf{L}\mathbf{z}$?

Family normal

$$\mathbb{E}(\boldsymbol{\mu} + \mathbf{L}\mathbf{z}) = \boldsymbol{\mu} + \mathbf{L}\mathbb{E}(\mathbf{z}) = \boldsymbol{\mu}$$

$$\text{Cov}(\boldsymbol{\mu} + \mathbf{L}\mathbf{z}) = \mathbf{L}\mathbf{I}\mathbf{L}' = \Sigma$$

Random generation from Normal

Example

Write a function to draw from $N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ where

$$\boldsymbol{\mu} = \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} 4 & 1 & 0 \\ 1 & 2 & -0.5 \\ 0 & -0.5 & 6 \end{pmatrix}.$$

Simulate many independent draws and verify that they are from the correct distribution.

Matrix inversion

Evaluating the PDF of MVN requires calculation of $\det(\Sigma)$ and $(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})$.

Warning

Calculating $(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})$ with

```
1 (x - mu) %*% solve(Sigma) %*% (x - mu)
```

can introduce significant rounding error.

It is more numerically stable, and thus preferred, to solve the related system of equations:

```
1 dev <- x - mu
2 Siginv_dev <- solve(Sigma, dev)
3 crossprod(dev, Siginv_dev)
```

where `crossprod(y, z)` calculates $\mathbf{t}(y) \%*\% z$.

Even better:

```
1 L <- t(chol(Sigma))
2 crossprod(forwardsolve(L, (x - mu)))
```

where `forwardsolve()` is designed for solving triangular systems.

Multivariate Student- t

Multivariate Student- t

Definition

Let $\boldsymbol{\mu} \in \mathbb{R}^D$, $\boldsymbol{\Sigma}$ be a $D \times D$ positive-definite matrix, and $\nu > 0$ be the degrees of freedom (usually an integer).

The location-scale **Multivariate Student- t** density for $\boldsymbol{x} = (x_1, \dots, x_D) \in \mathbb{R}^D$ is

$$f(\boldsymbol{x}) = \frac{\Gamma((D + \nu)/2)}{\Gamma(\nu/2) (\pi\nu)^{D/2}} \det(\boldsymbol{\Sigma})^{-1/2} \left[1 + \frac{1}{\nu} (\boldsymbol{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\boldsymbol{x} - \boldsymbol{\mu}) \right]^{-\frac{D+\nu}{2}}.$$

We write $\boldsymbol{x} \sim t_D(\nu, \boldsymbol{\mu}, \boldsymbol{\Sigma})$.

- $\mathbb{E}(\boldsymbol{x}) = \boldsymbol{\mu}$ if $\nu > 1$.
- $\text{Cov}(x_i, x_j) = \frac{\nu}{\nu-2} (\boldsymbol{\Sigma})_{i,j}$ if $\nu > 2$.
- Heavier tails than Multivariate Normal.

Wishart and Inverse-Wishart

Wishart distribution

Definition

Let Ψ^{-1} be a $D \times D$ positive-definite scale matrix and $\nu > 0$ be the degrees of freedom (usually an integer).

The **Wishart** density for random positive definite matrix Σ^{-1} is

$$f(\Sigma^{-1}) \propto \det(\Sigma^{-1})^{(\nu-D-1)/2} \exp \left[-\frac{1}{2} \text{tr}(\Psi \Sigma^{-1}) \right].$$

We write $\Sigma^{-1} \sim \text{Wish}(\nu, \Psi^{-1})$.
 Ψ^{-1} is the scale matrix

- $\mathbb{E}(\Sigma^{-1}) = \nu \Psi^{-1}$
- Multivariate version of Gamma (χ^2) distribution
- Conditionally conjugate to the *precision matrix* in Multivariate Normal
- See [Wishart Wikipedia page](#) for properties, including generating random matrices

Inverse-Wishart distribution

Definition

Let Ψ be a $D \times D$ positive-definite scale matrix and $\nu > 0$ be the degrees of freedom (usually an integer).

The **Inverse-Wishart** density for random positive definite matrix Σ is

$$f(\Sigma) \propto \det(\Sigma)^{-(\nu+D+1)/2} \exp \left[-\frac{1}{2} \text{tr}(\Psi \Sigma^{-1}) \right].$$

We write $\Sigma \sim \text{Inv-Wish}(\nu, \Psi)$.

- If $\Sigma^{-1} \sim \text{Wish}(\nu, \Psi^{-1})$, then $\Sigma \sim \text{Inv-Wish}(\nu, \Psi)$
- $\mathbb{E}(\Sigma) = \frac{1}{\nu-D-1} \Psi$
- Conditionally conjugate to the *covariance matrix* in Multivariate Normal
- See [Inverse-Wishart Wikipedia page](#) for properties, including generating random matrices

Modeling with Multivariate Normal

Multivariate Normal likelihood

Proposition

Let $\mathbf{y}_i \stackrel{\text{iid}}{\sim} \text{Normal}_D(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ for $i = 1, \dots, n$.

The likelihood is given as

$$(2\pi)^{-nD/2} \det(\boldsymbol{\Sigma})^{-n/2} \exp \left[-\frac{n-1}{2} \text{tr}(\boldsymbol{\Sigma}^{-1} \mathbf{S}) - \frac{n}{2} (\bar{\mathbf{y}} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\bar{\mathbf{y}} - \boldsymbol{\mu}) \right],$$

where $\bar{\mathbf{y}} = \frac{1}{n} \sum_{i=1}^n \mathbf{y}_i$,

$\mathbf{S} = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{y}_i - \bar{\mathbf{y}})(\mathbf{y}_i - \bar{\mathbf{y}})^\top$ is a $D \times D$ estimate of the covariance matrix, and

$\text{tr}(\mathbf{X}) = \sum_j (\mathbf{X})_{j,j}$ is the trace function.

Proof:

$$\begin{aligned} f(\mathbf{y}) &= \prod_{i=1}^n (2\pi)^{-D/2} \det(\boldsymbol{\Sigma})^{-1/2} \exp \left[-\frac{1}{2} (\mathbf{y}_i - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{y}_i - \boldsymbol{\mu}) \right] \\ &= (2\pi)^{-nD/2} \det(\boldsymbol{\Sigma})^{-n/2} \exp \left[-\frac{1}{2} \sum_{i=1}^n (\mathbf{y}_i - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{y}_i - \boldsymbol{\mu}) \right] \\ Q &= \sum_{i=1}^n (\mathbf{y}_i - \bar{\mathbf{y}} + \bar{\mathbf{y}} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{y}_i - \bar{\mathbf{y}} + \bar{\mathbf{y}} - \boldsymbol{\mu}) = \sum_{i=1}^n (\mathbf{y}_i - \bar{\mathbf{y}})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{y}_i - \bar{\mathbf{y}}) + \end{aligned}$$

$$+ \sum_i (\bar{y} - \mu)^T \Sigma^{-1} (\bar{y} - \mu)$$

$$\sum_{i=1}^n (y_i - \bar{y})^T \Sigma^{-1} (y_i - \bar{y})$$

$1 \times D$ $D \times D$ $D \times 1$

$$= \sum \text{tr}((y_i - \bar{y})^T \Sigma^{-1} (y_i - \bar{y}))$$

$$= \sum \text{tr}(\Sigma^{-1} (y_i - \bar{y}) (y_i - \bar{y})^T)$$

$$= \text{tr}(\Sigma^{-1} \underbrace{\sum_i (y_i - \bar{y}) (y_i - \bar{y})^T}_{\text{Sample "variance"}})$$

MVN model: fully conjugate prior

Normal-Inverse-Wishart

Let $\pi(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \pi(\boldsymbol{\Sigma}) \times \pi(\boldsymbol{\mu} \mid \boldsymbol{\Sigma})$

with

$$\begin{aligned}\boldsymbol{\Sigma} &\sim \text{Inv-Wish}(n_0, n_0 \boldsymbol{S}_0), \\ \boldsymbol{\mu} \mid \boldsymbol{\Sigma} &\sim \text{Normal}_D\left(\boldsymbol{m}_0, \frac{1}{\kappa_0} \boldsymbol{\Sigma}\right).\end{aligned}$$

- See [Normal-inverse-Wishart distribution \(Wikipedia\)](#)
- $\boldsymbol{S}_0$ is the prior (matrix) harmonic mean of $\boldsymbol{\Sigma}$, i.e., $[\mathbb{E}(\boldsymbol{\Sigma}^{-1})]^{-1}$

MVN model: fully conjugate prior

Prior

$$\pi(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \text{Inv-Wish}(\boldsymbol{\Sigma}; n_0, n_0 \mathbf{S}_0) \times \text{Norm}_D(\boldsymbol{\mu}; \mathbf{m}_0, \frac{1}{\kappa_0} \boldsymbol{\Sigma})$$

n_0 : prior sample size for $\boldsymbol{\Sigma}$

$\mathbf{S}_0$: prior guess for $\boldsymbol{\Sigma}$

$\mathbf{m}_0$: mean of prior for $\boldsymbol{\mu}$

κ_0 : prior sample size for $\boldsymbol{\mu} | \boldsymbol{\Sigma}$

MVN model: fully conjugate prior

Posterior

$$\pi(\boldsymbol{\mu}, \boldsymbol{\Sigma} \mid \{\mathbf{y}_i\}) = \text{Inv-Wish}(\boldsymbol{\Sigma}; n_1, n_1 \mathbf{S}_1) \times \text{Norm}_D(\boldsymbol{\mu}; \mathbf{m}_1, \frac{1}{\kappa_1} \boldsymbol{\Sigma})$$

$$n_1 =$$

$$n_1 \mathbf{S}_1 =$$

$$\mathbf{m}_1 =$$

$$\kappa_1 =$$

MVN model: fully conjugate prior

Posterior

$$\pi(\boldsymbol{\mu}, \boldsymbol{\Sigma} \mid \{\mathbf{y}_i\}) = \text{Inv-Wish}(\boldsymbol{\Sigma}; n_1, n_1 \mathbf{S}_1) \times \text{Norm}_D(\boldsymbol{\mu}; \mathbf{m}_1, \frac{1}{\kappa_1} \boldsymbol{\Sigma})$$

- $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ can be sampled sequentially with the compositional method.
- The marginal posterior of $\boldsymbol{\mu}$ is

$$t_D \left(\nu_1, \mathbf{m}_1, \frac{1}{\kappa_1 \nu_1} n_1 \mathbf{S}_1 \right)$$

where $\nu_1 = n_1 - D + 1$.

MVN model: fully conjugate prior

Posterior predictive

$$f(\tilde{\mathbf{y}} \mid \{\mathbf{y}_i\}) = \int \int f(\tilde{\mathbf{y}} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) \pi(\boldsymbol{\mu}, \boldsymbol{\Sigma} \mid \{\mathbf{y}_i\}) d\boldsymbol{\mu} d\boldsymbol{\Sigma}$$

which is

$$t_D \left(\nu_1, \mathbf{m}_1, \frac{\kappa_1 + 1}{\kappa_1 \nu_1} n_1 \mathbf{S}_1 \right)$$

where $\nu_1 = n_1 - D + 1$.

How could you sample from this posterior predictive distribution?

MVN model: conditionally conjugate prior

Conditionally conjugate prior

$$\pi(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \text{Normal}_D(\boldsymbol{\mu}; \boldsymbol{m}_0, \boldsymbol{V}_0) \times \text{Inv-Wish}(\boldsymbol{\Sigma}; \nu_0, \boldsymbol{\Psi}_0)$$

Posterior **complete (or full) conditionals**

$\boldsymbol{\mu} \mid \boldsymbol{\Sigma}, \{\boldsymbol{y}_i\} \sim \text{Normal}_D(\boldsymbol{m}_1, \boldsymbol{V}_1)$ with

$$\boldsymbol{V}_1 = (\boldsymbol{V}_0^{-1} + n\boldsymbol{\Sigma}^{-1})^{-1} \quad \text{and} \quad \boldsymbol{m}_1 = \boldsymbol{V}_1 (\boldsymbol{V}_0^{-1}\boldsymbol{m}_0 + n\boldsymbol{\Sigma}^{-1}\bar{\boldsymbol{y}}),$$

and

$\boldsymbol{\Sigma} \mid \boldsymbol{\mu}, \{\boldsymbol{y}_i\} \sim \text{Inv-Wish}(\nu_1, \boldsymbol{\Psi}_1)$ with

$$\nu_1 = \nu_0 + n \quad \text{and} \quad \boldsymbol{\Psi}_1 = \boldsymbol{\Psi}_0 + \sum_{i=1}^n (\boldsymbol{y}_i - \boldsymbol{\mu})(\boldsymbol{y}_i - \boldsymbol{\mu})^\top.$$

Proof:

$$\pi(\mu) = N(m_0, V_0)$$

$$\text{density} \propto \exp\left[-\frac{1}{2}Q\right]$$

$$\text{with } Q = (\mu - m_0)^T V_0^{-1} (\mu - m_0)$$

$$= \mu^T V_0^{-1} \mu - \mu^T V_0^{-1} m_0$$

$$- m_0^T V_0^{-1} \mu + m_0^T V_0^{-1} m_0$$

$$= \mu^T V_0^{-1} \mu - 2m_0^T V_0^{-1} \mu + \text{const}$$

$$\text{Likelihood} \propto \exp\left[-\frac{1}{2}Q_L\right]$$

$$Q_L = n(\bar{y} - \mu)^T \Sigma^{-1} (\bar{y} - \mu)$$

Combine Lik x Prior

$$\pi(\mu | y, \Sigma) \propto \exp\left[-\frac{1}{2} Q_1\right]$$

$$Q_1 = \mu^T V_0^{-1} \mu - 2 m_0^T V_0^{-1} \mu + \text{const} \\ + n \mu^T \Sigma^{-1} \mu - 2 \bar{y}_n^T \Sigma^{-1} \mu + \text{const}_2$$

$$= \mu^T \underbrace{[n \Sigma^{-1} + V_0^{-1}]}_{V_1^{-1}} \mu + \dots \text{getting close!}$$

our goal is to get to something that looks like Q in blue above.

MVN model



Tip

When working with the fully and conditionally conjugate models, computing from the equivalent perspective of Σ^{-1} avoids most matrix inversions.

MVN model

Alternative models for Multivariate Normal data

- Noninformative prior (see Ch 3.6 of BDA3)
- Jeffreys prior (see Ch 3.6 of BDA3)
- Scaled Inverse-Wishart (see Ch 3.6 of BDA3)
- Square-root-free Cholesky factorization of [Daniels and Pourahmadi \(2002\)](#)
 - $\Sigma = \mathbf{B}^{-1} \mathbf{D} (\mathbf{B}^{-1})^\top$ where $\mathbf{D}$ is diagonal with independent Inverse-Gamma entries and $\mathbf{B}$ is unit-lower-triangular with independent Normal entries.
 - Conditionally conjugate
 - More flexible and intuitive (regression interpretation) than Wishart

Palmer penguins

Example

The `penguins` data set in the `palmerpenguins` R package contains measurements on 344 penguins among three different species in the Palmer Archipelago in Antarctica.

1. Perform an EDA of bill length, bill depth, flipper length, and body mass for females of the Adelie and Chinstrap species.
2. Propose a multivariate normal model for these variables and sample the posterior distribution of the parameters.
3. Compare bill-length and flipper-length mean profiles between the two species (univariate and bivariate).

